

Mid Examination

Semester 20252

Subject : **Web Application Development**

Study Program : Informatics / Information Systems

Student Name :

Student ID :

Instructions for Students

1. This examination consists of 3 questions.
2. All answers to be written in answer using Ms. Word, submitted by pdf.
3. Students are to use proper English and are required to write neatly and clearly.
4. Students are allowed to use dictionaries and/or calculators if needed
5. Any communication devices are prohibited
6. Attempting communication between fellow students are prohibited
7. Any attempts on cheating and plagiarism will result in an immediate zero score..

Course Learning Outcomes

CLO 1	Students are able to analyze and identify the needs of designing server side web applications to provide solution in a cross-disciplinary fields
CLO 2	Students are able to design, implement, and evaluate server side web application based on a certain set of requirements
CLO 3	Students are able to apply the theories and principles of server side web programming to deliver a solution based on certain set of requirements

1. As technology usage increases, every aspect in the world is affected, including the human resources. Based on the information relayed in the government of the Republic of Indonesia, it is necessary for a working citizen to have a Tax Identification Number or NPWP. To make things easier, you are tasked with ***analyzing and identifying the needs to create a helper application to calculate the tax***. The requirements are as follows.
- b. A citizen could enter his/her tax amount (must be positive and must be a number)
 - c. If a citizen has NPWP, the tax entered is not added with anything
 - d. If a citizen does not have NPWP, the tax will be added with extra 20% of the initial tax amount
 - e. Analyze and Identify the needs to get and show the tax amount, then provide the code along with the HTML

Note: you could use this image as the base benchmark for HTML form view.

☐ Has NPWP

☐ Has NPWP

Tax amount: 120000

☒ Has NPWP

Tax amount: 100000

2. You are working in a software company, and being tasked with analyzing the needs to create a simple application to design a 3x3 grid containing the numbers as seen in the following image. **Analyze and provide the code necessary to produce the same output. You must use `range()` function to do this.**

1	2	3
4	5	6
7	8	9

3. A shop wants your help to design a table containing grocery data. The shop owner wants you to show the subtotal and grand total of the grocery as in the following table.

Name	Qty	Price	Subtotal
Milk	2	5000	10000
Tofu	4	10000	40000
Tea	3	3000	9000
Candy	5	1000	5000
Cookies	4	4500	18000
Total			82000

You may use the following data as the starting point

```
$data = [  
  ["name" => "Milk", "qty" => 2, "price" => 5000],  
  ["name" => "Tofu", "qty" => 4, "price" => 10000],  
  ["name" => "Tea", "qty" => 3, "price" => 3000],  
  ["name" => "Candy", "qty" => 5, "price" => 1000],  
  ["name" => "Cookies", "qty" => 4, "price" => 4500],  
];
```

Construct the output based on the provided data.